Onyx
Touchpad maker Synaptics' prototype TS phone Onyx distinguishes touch based on the area of contact

Apple again
"Multitouch" is registered as a trademark by Apple Computer, Inc.

and bottom layer to find the X and Y coordinates of the point of contact. An 8 wire Resistive interface is similar to the 4 wire, but adds 4 more electrodes to improve the accuracy in TSs larger than 10 inches (diagonal) and also counter the effects of drift - a phenomenon where over time the flexible film shifts position causing inaccurate readings.

A slight variation in the 4 wire design is the 5 wire design which has 4 electrodes on the bottom layer - one each corner - and one on the top flexible layer. This design increases the screen's durability and accuracy. A further refinement of this design is the 6 wire interface which uses one additional ground wire to reduce interference and increase accuracy. A 7 wire interface adds two additional wires to the 5 wire design to reduce effect of drift.

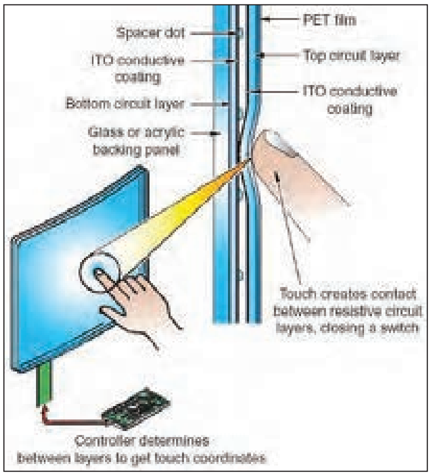
Resistive TS require some pressure to be applied to make the two layers meet, mere touch doesn't suffice, and this explains your encounter with the TS at the ATM. And this issue aggravates as the ITO layer gets worn with use.

Most PDAs that feature a stylus use this technology. Also seen in some mobile phones, notably the Motorola ROKR. Due to the flexible top layer of the interface it is easy to spot a Resistive TS.

Advantages
Cheap to make. Any object can be used to make contact. Accuracy is high, so this interface is ideal for note taking and character recognition activities.

Disadvantages
Top layer is susceptible to damage. The interface is not usable once the top layer is damaged. Since there are many layers piled upon the actual display, visibility is reduced. Usually around 75 per cent transmission is promised. Cannot sense multiple points of contact Wear can result in the ITO coating to become less sensitive, therefore over time accuracy decreases and recalibration is needed.

Capacitive TS
Capacitive TS can be of two types - Surface Capacitive and Projected Capacitive. In both cases, the top layer is usually



Resistive touchscreens rely on variation in resistivity

glass making it more scratch resistant than Resistive TS. The core property used here is that of a capacitor - two conductors separated by an insulator.

In Surface Capacitive screens, the glass is coated with a conductive layer, usually Indium Tin Oxide, on the underside. A small amount of voltage is applied to uniformly charge the entire surface. There are electrodes on the four corners of the screen to check for voltage changes. When a conducting object, like a human finger or any metal object held by the human hand, touches the screen, the charge is drained at that point (the glass acts as the insulator between the conductive layer and the conductive object), and a voltage change is registered. Each electrode detects this change, to arrive at the point of contact.

Projected Capacitive screens consist of two parallel layers of conducting material (ITO), similar to the set-up seen in Resistive TS. The difference is that there are no flexible layers - both layers are glass, and the conducting material is etched into their proximal surfaces so as to form a grid of electrodes separated by a film of air (thereby creating a grid of capacitors). The capacitance field is active above the surface of the top layer of glass, and bringing a conductive material like a human finger even close (2 mm is enough) to the touchscreen causes a discharge and resultant voltage change which is picked up by the electrodes. The iPhone and many premium smartphones use this technology.

Advantages
True touchscreens, where mere touch generates input. Projected Capacitive screens allow detection of multiple points of contact, allowing multi touch operations/gestures. Compared to Resistive screens, these screens allow faster inputs. Since glass isn't easily scratched, this is more rugged than resistive TS.

Disadvantages
Needs a conducting object to register touch, so ordinary stylus or even a finger nail will not do. Scratches on the top layer can impact performance.

Surface acoustic wave TS
This technology employs ultrasonic waves (USW) to form a grid above the screen. Piezoelectric USW generators and reflectors are used to generate US waves and stream them over the surface - usually glass. US waves are generated on both vertical and horizontal directions by generators, while receivers placed at the opposite side monitor them. When a finger or object is placed on the screen or near it, some of the US waves are blocked. This is detected by the receivers in both axes, and the location of the point of touch is calculated.

Advantages
Since there are no layers involved, just a glass sheet, visibility is very high. Works with any object. The only TS technology that can also detect Z axis or pressure changes if any.

Disadvantages
Since the network of US waves is above the surface, it is possible that false touches are registered if objects come close. This doesn't bedevil Guided Acoustic wave screens. Expensive.

Scanning infrared TS
Similar in concept and function as Surface Acoustic Wave TS, here IR diodes and sensors are placed on opposite sides of the bezel to generate and receive IR waves respectively. The obstruction in the IR waves at the point where the object is